
Early Warning System for Telco Customer Churn: Predicting At-Risk Customers and Prioritizing Retention

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Abstract

This project proposes an early warning system for customer churn in the telecommunications sector. Using the IBM Telco Customer Churn dataset, the study aims to predict which customers are at risk of churning and identify the most influential drivers behind that behavior. The project compares logistic regression, random forest, and gradient boosting models to balance interpretability and predictive performance, while using evaluation metrics such as precision, recall, F1 score, and AUC ROC. The expected outcome is a practical framework for prioritizing retention efforts and generating actionable business insights.

1 Research Question and Motivation

Customer churn is a major challenge for subscription-based and service-oriented businesses because losing existing customers is often more costly than retaining them. Recognizing the importance of your efforts in this area, companies in industries such as telecommunications, banking, and software services rely heavily on stable customer relationships, making churn prediction an important data science problem with direct financial consequences. If organizations can identify which customers are most likely to leave before churn occurs, they can take targeted actions such as offering discounts, improving service, or prioritizing customer support.

The main research question of this project is: Can we accurately predict which customers are likely to churn and identify the most influential factors? Highlighting this practical aspect helps readers grasp the study's real-world impact, making the methodology and findings more compelling.

This question is interesting because it combines predictive modeling, interpretability, and business strategy. Prior work in customer analytics has shown that variables such as contract type, tenure, monthly charges, and service usage often influence churn behavior. However, different models vary in both predictive power and interpretability. Therefore, comparing simpler, interpretable methods such as logistic regression with more complex machine learning models, such as random forests and gradient boosting, can provide both analytical rigor and practical value.

2 Data Sources

This project will use a public customer churn dataset, with the IBM Telco Customer Churn dataset found on Kaggle serving as the primary candidate; the dataset can be viewed [here](#). This dataset contains 7,043 customer records and 33 variables describing demographics, account characteristics, subscribed services, billing behavior, and churn outcomes. The data will be obtained directly from the uploaded spreadsheet and imported into Python for analysis using Pandas.

During cleaning and preprocessing, we will remove/ignore non-informative identifiers or constant fields such as CustomerID, Count, Country, and State, convert Total Charges from text to numeric format, and handle the small number of blank values in that column, which appear mainly for

customers with zero tenure. We will also encode categorical variables such as Contract, Internet Service, and Payment Method into standardized numerical variables if needed for models such as logistic regression.

A key preprocessing concern is target leakage, as columns such as Churn Reason and Churn Score directly reflect churn outcomes and cannot therefore be used as predictors in the model. Additional challenges include deciding whether to exclude high-cardinality geographic variables such as City and Zip Code, and addressing any class imbalance in the churn label. Overall, this dataset is appropriate for the full project because it is large, structured, and already contains a clear target variable (Churn Value) along with enough customer-level features to support exploratory analysis, predictive modeling, and interpretable business recommendations.

3 Methodology and Analysis Plan

For this project, we decided to create a classification model. Since the dataset has 33 variables, we will start with logistic regression as our baseline due to its interpretability, and then compare it with additional classifiers, such as Random Forest and Gradient Boosting, to evaluate trade-offs between accuracy and explainability. We will use AIC for logistic regression model comparison and feature importance scores from tree-based models to identify the most influential variables while reducing noise and multicollinearity among highly correlated predictors. Using AIC will also allow us to remove non-significant variables from the model.

We plan to use NumPy and Pandas to create a data frame, identify outliers ($\pm 3\sigma$), and handle missing values based on the variable: we might drop the entire entry, fill it with the column mean, or replace it with 0. Once the models are fit, we will evaluate performance using a Confusion Matrix, and address potential class imbalance in the churn label by also reporting Precision and Recall. For feature selection, we will use AIC alongside F-tests to compare models and drop insignificant variables, and Chi-Squared tests to assess associations between categorical predictors and the churn label. We will also test for overfitting by comparing training and validation performance. Some of the libraries we will use are: NumPy, Pandas, Scikit-Learn, Matplotlib, and Seaborn.

4 Expected Outcomes and Evaluation

Contract type, tenure, monthly charges, and internet service type will be seen as the key drivers of customer churn by these models. With a 26.5% churn rate in this dataset and a large proportion of month-to-month contracts, we anticipate that customers with shorter tenure and higher monthly charges will be seen as at higher risk of churning by these models. We also anticipate that customers not purchasing add-on services such as online security or tech support will be seen as at higher risk of churn, which could represent a potential strategy for reducing customer churn.

To evaluate the model, we'll consider a train-test split (80/20) and use the confusion matrix, precision, recall, F1 score, and AUC ROC as evaluation metrics. Due to the moderate level of class imbalance in the dataset, we'll not consider the accuracy metric for evaluation, and instead, we'll emphasize the recall of the churn class to ensure that we don't miss any customers who are at risk of leaving the company. Based on the evaluation metrics, we'll compare logistic regression, random forest, and gradient boosting models to determine the trade-off between model interpretability and performance. The logistic regression model is expected to perform well in terms of model interpretability, while the gradient boosting model is expected to perform better in terms of AUC ROC evaluation metrics. To avoid overfitting, we'll compare the performance of the model on the training and validation sets and consider k-fold cross-validation for model evaluation.

Other than the accuracy of the models, we also want to extract feature importance rankings, as well as, if time allows, compute SHAP values for further understanding of the predictions being generated. This will enable us to take the results of the models and use them to generate actionable business decisions, such as determining which customer profiles to focus on for retention efforts. Another extension of this project could be to create a basic dashboard for risk scoring, which will alert customers with high probabilities of churning, thus closing the loop between modeling and decision-making.